

2024 AMC 10A Problem 8 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 8



Problem:

Amy, Bomani, Charlie, and Daria work in a chocolate factory. On Monday Amy, Bomani, and Charlie started working at 1:00 PM and were able to pack 4, 3, and 3 packages, respectively, every 3 minutes. At some later time, Daria joined the group, and Daria was able to pack 5 packages every 4 minutes. Together, they finished packing 450 packages at exactly 2 : 45PM. At what time did Daria join the group?

Answer Choices:

- A. 1 : 25PM
- B. 1 : 35PM
- C. 1 : 45PM
- D. 1 : 55PM
- E. 2 : 05PM

Solution:

Every 3 minutes, Amy, Bomani, and Charlie together packed 10 packages. From 1:00 PM to 2 : 45PM, a span of $60 + 45 = 105$ minutes, these three packers packed $\frac{105}{3} \cdot 10 = 350$ packages. This means that Daria must have packed $450 - 350 = 100$ packages. The time needed for Daria to pack 100 packages is $\frac{100}{5} \cdot 4 = 80$ minutes. Therefore Daria joined the group at (B) 1 : 25PM, which is 80 minutes before 2 : 45PM.

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